

Prakhar Singh

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RESEARCH PROFILE

Thesis: Solar Eruptions Across Different Scales: Energetics and Coupled Dynamics

My research investigates the energetics and coupled dynamics of solar eruptive phenomena, primarily flares and coronal jets across a range of spatial and temporal scales, using coordinated multi-instrument observations from SDO, IRIS, Hinode, Solar Orbiter, and Aditya-L1. Combining imaging, spectroscopic diagnostics, and magnetic field extrapolation, I probe the thermal, kinematic, and magnetic signatures of these eruptions to understand their role in mass loading of the solar wind and coronal heating, as well as their broader implications for space-weather forecasting.

EDUCATION

Doctor of Philosophy (ongoing)

Aryabhata Research Institute of Observational Sciences (ARIES)

Affiliation: Indian Institute of Technology (IIT) Roorkee

Dec 2023 – Present

Master of Science, Physics

Banaras Hindu University — CGPA: 8.7/10

2021 – 2023

Bachelor of Science, Physics

Aligarh Muslim University — CGPA: 8.1/10

2018 – 2021

PUBLICATIONS (in preparation)

Manuscripts in preparation for peer-reviewed submission.

- Singh, P., et al. X-ray Spectroscopic Diagnostics of M-Class Solar Flares using SoLEXS onboard Aditya-L1 (submitted to *Journal of Astrophysics and Astronomy (Topical Issue)*)
- Singh, P., et al. From Standard to Blowout: Plasmoid-Driven Morphology Transition of a Recurrent Solar Jet (In preparation for submission to the *Astrophysical Journal (APJ)*)

RESEARCH EXPERIENCE

Spectral Diagnostics of Confined and Eruptive Solar Flares Using SoLEXS/Aditya L1

July 2025 – Present

- Investigating confined and eruptive solar flares through spectral analysis using SoLEXS, characterizing thermal and compositional differences between the two flare classes.

X-ray Spectral Fitting Pipeline for SoLEXS and HELIOS onboard Aditya-L1

- Developed a X-ray spectral-fitting pipeline (XSPEC, chisoth isothermal CHIANTI model) to derive plasma temperature, emission measure, and elemental abundances for solar flares.
- Developed a joint spectral-fitting pipeline combining SoLEXS and HELIOS data, integrating soft and hard X-ray diagnostics to constrain flare plasma properties across a broader energy range.

Energetic and Kinematic Properties of Recurrent Solar Active-Region Jets

Jan – May 2025

- Characterized a recurrent blowout-jet event using SDO/AIA, HMI, IRIS and Hinode/XRT identifying a plasmoid-driven transition from standard to blowout jet morphology.
- Derived jet kinematics (length, width, duration, velocity) and, DEM analysis to derive emission measure, temperature, and electron density; estimated kinetic, potential, enthalpy, and radiative energy-flux components.
- Performed non-force-free field (NFFF) magnetic extrapolation to reveal a null-point underlying the jet, and associated the eruption with a Type III radio burst.

Chromospheric Heating During Active-Region Evolution

Jun – Dec 2023

Manipal Centre for Natural Sciences, MAHE — Supervisor: Dr. Sreejith Padinhatteeri

- Examined the correlation between an active region's UV/EUV emission and photospheric magnetic activity using HMI and AIA channels.
- Derived temperature and emission-measure maps using differential emission measure (DEM) analysis.

M.Sc. Thesis — Supervisor: Prof. Abhay Kumar Singh, Banaras Hindu University

- Investigated nighttime D-layer height, electron density, and propagation distance using Very Low Frequency (VLF) data from an Automatic Whistler Detector network.
- Characterized diurnal and seasonal variability of tweek occurrence over the South Asian region.

TECHNICAL SKILLS

Missions & Instruments: SDO (AIA, HMI), IRIS, Hinode (XRT), Solar Orbiter (EUI, STIX, PHI), Aditya-L1 (SoLEXS, HEL1OS, SUIT), Chandrayaan-2 (XSM), e-CALLISTO, GOES XRS

Analysis Techniques: X-ray spectroscopic diagnostics, differential emission measure (DEM) analysis, non-force-free field (NFFF) magnetic extrapolation, time-series/wavelet analysis, image processing and height-time analysis

Programming & Computing: Python, IDL/SolarSoft (SSW), Shell scripting, Git

Scientific Software: XSPEC, Sherpa, VAPOR, SAOImage DS9, LaTeX

CONFERENCE TALKS & POSTERS

Contributed Talk

Confined vs. Eruptive Solar Flares: A Thermal and Compositional Diagnostic Study with SoLEXS

ASI Symposium 003 - Cosmic Vision 2047: Solar and Planetary Dynamics through Observations and AI/ML, JECRC University, Jaipur 8–10 Sep 2025

Poster Presentations

X-ray Spectroscopic Diagnostics of Confined and Eruptive Flares Using SoLEXS/Aditya-L1

International Conference on Exploring the Sun at High-Resolution: Present Perspectives and Future Horizons, Udaipur 10–13 Feb 2026

Diagnosing Solar Flare Dynamics: A Comparative Study of Confined and Eruptive Events Using SoLEXS/Aditya-L1

ISRO-ESA Heliophysics Workshop on Aditya-L1, Solar Orbiter and Proba-3, Thiruvananthapuram 19–23 Jan 2026

Thermal and Compositional Signatures of Confined and Eruptive Flares with SoLEXS/Aditya-L1

The Variable Sun: Past, Present and Future, VSSC, Thiruvananthapuram 13–17 Oct 2025

Energetic and Kinematic Properties of Recurrent Solar Active Region Jets: Multi-Wavelength Analysis with AIA and IRIS Observations

Sun, Space Weather and Solar-Stellar Connections, IIA, Bengaluru 20–24 Jan 2025

CONFERENCE/WORKSHOP/SCHOOL ATTENDED

- Attended the STIX Team Meeting 2026 (online) 22 – 24 July 2026
- 2nd European Solar Physics Division (ESPD) Summer School (online) 27 Apr – 1 May 2026
- Mentored a project on SoLEXS data analysis in ATSOAA 2026, ARIES, Nainital 20–30 Apr 2026
- Solar Orbiter Hackathon, ESA/ESTEC, Netherlands — Team 3 (online) 17–21 Nov 2025
- National-Level School on Analysis and Statistical Modelling of Space Science Data, MAHE 7–12 Jul 2025
- Solar Cycle Variability Conference 2024, IIT BHU & ARIES, Nainital 14–18 Oct 2024
- 2nd Workshop on Space Weather Science and Opportunities, IIT Roorkee 5–9 Oct 2024
- Design of Space Payloads and Developmental Approach, MAHE Jun 2023
- IMPRESS 2023, Indian Institute of Geomagnetism 11–14 Feb 2023
- 11th IIST Astronomy and Astrophysics School, IIST 14–23 Dec 2022
- 1st Indo-US Collaborative Meeting Space Radiation Workshop, ARIES 24–28 Jan 2022

Aditya-L1 Support Cell Workshops

Instructor / Mentor

- 13th Workshop, ARIES, Nainital — guided SoLEXS data-analysis project 6–15 Jul 2026
- 11th Workshop, Pondicherry University — guided SoLEXS data-analysis project 6–9 Oct 2025
- 10th Workshop, ARIES, Nainital — guided SoLEXS data-analysis project 26 May – 4 Jun 2025

Participant

- 9th Workshop, IIT Guwahati 24–26 Mar 2025
- 7th Workshop, ARIES, Nainital 21–30 May 2024
- 5th Workshop, IIT Kanpur 29 Sep – 1 Oct 2023
- 3rd Workshop, IIT BHU 25–27 Feb 2023

RESEARCH VISITS

Udaipur Solar Observatory (USO), Udaipur, India

Feb 15 – Mar 20, 2026

ISRO RESPOND Project — developed a joint spectral-fitting pipeline for SoLEXS and HELIOS onboard Aditya-L1.

Udaipur Solar Observatory (USO), Udaipur, India

Sep 11–17, 2025

ISRO RESPOND Project — trained in non-force-free field (NFFF) extrapolation for coronal magnetic field reconstruction.

Udaipur Solar Observatory (USO), Udaipur, India

Mar 6–20, 2025

ISRO RESPOND Project — Chandrayaan-2/XSM data processing, calibration, and spectral interpretation.

Kodaikanal Solar Observatory (KSO), Kodaikanal, India

Jan 25–27, 2025

Familiarization with historical and modern solar observation methods and instrumentation.

AWARDS & HONOURS

- 3rd rank, M.Sc. (Physics), Banaras Hindu University, class of 2023.
- Top 5%, *Introduction to Research* (NPTEL), IIT Madras.
- 4th rank, *Electronic Theory of Solids* (NPTEL), IIT Kharagpur.
- University topper, National Graduate Physics Examination (NGPE-2021), Indian Association of Physics Teachers.
- Student Ambassador, International Astronomy and Astrophysics Competition (IAAC-20).